

Studentkare — System Architecture & Flow Specification

Enterprise Healthcare Platform • Tata 1mg AI Ecosystem • ABDM Vault Standard

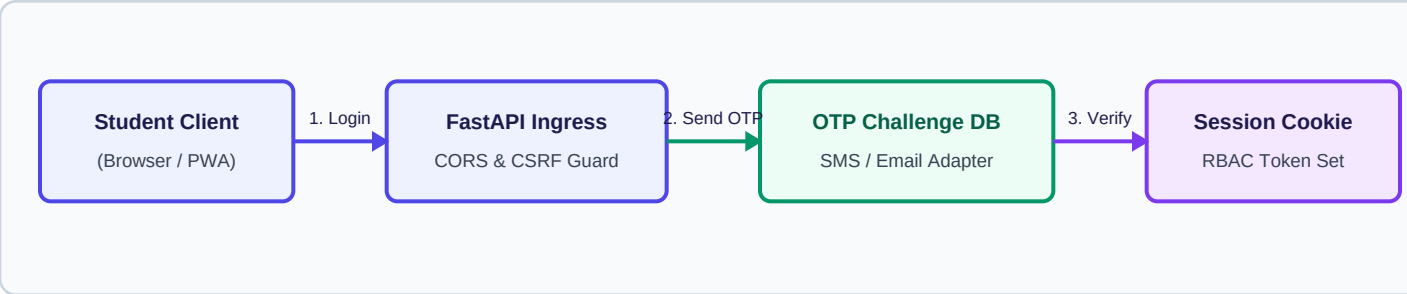
1. System Architecture Overview

Studentkare is designed as a high-concurrency, security-first student health management platform. It connects React 18 frontend workspaces with an asynchronous FastAPI microservices backend, autonomous AI agent clusters, Tata 1mg healthcare catalog services, and ABDM (Ayushman Bharat Digital Mission) FHIR health vaults.

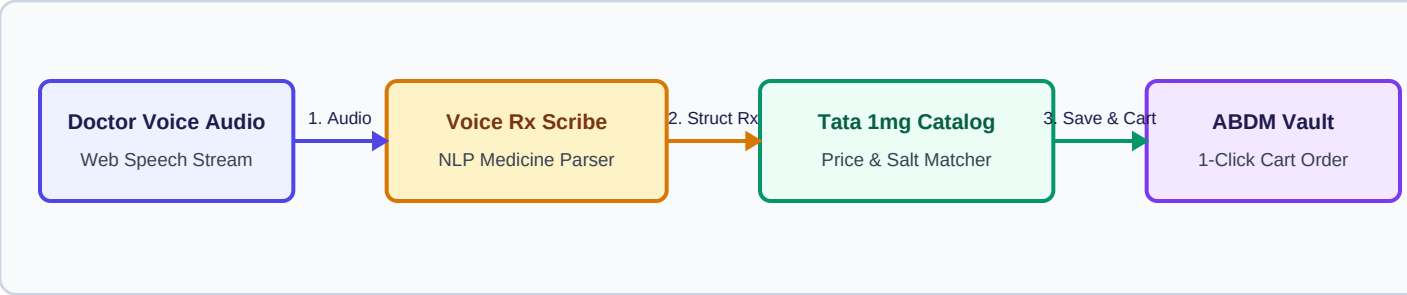
Layer	Technologies & Frameworks	Key Responsibilities
Presentation Layer	React 18, Vite, TypeScript 5.7, Lucide Icons, Impilo Pearl CSS	Student workspace, Tata 1mg store, AI Voice Rx modal, Admin console
API Services Layer	FastAPI, Python 3.11/3.14, Pydantic V2, Starlette Middleware	Async REST endpoints, session security, OTP verification, RBAC guard
AI Agents Subsystem	Multi-Doctor Swarm, Whisper Voice Scribe, PyTorch Acoustics	Triage synthesis, SOAP notes generation, prescription OCR, voice acoustics
Data & Cache Layer	PostgreSQL 16 (SQLAlchemy), Redis 7, Celery Workers, Firestore	Relational health storage, read-through cache, async task processing, realtime GPS
Integrations Layer	Tata 1mg M2M API, ABDM ABHA Gateway, Twilio, PostHog SDK	Catalog sync, 1-click cart checkout, ABHA FHIR records, FCM notifications

2. Core Operational Flow Diagrams

Flow 1: User Session, Dual-Channel OTP & CSRF Security Lifecycle



Flow 2: AI Voice Prescription Scribe & Tata 1mg Catalog Cart Sync



3. Core Data Storage Schema (`care\_\*` Relational Model)

Table Name	Primary Keys & Indexes	Key Columns & Encrypted Attributes
care_users	id (PK) IDX: (email, phone)	full_name, role, password_hash, is_active, abha_id
care_health_records	id (PK) IDX: (account_id, created_at)	title, category, mime_type, content (AES-256), fhir_bundle
care_catalog_items	id (PK) GIN: name_trgm	name, brand, active_chemical, price_paise, requires_rx
care_orders	id (PK) IDX: (user_id, status)	items_json, total_price_paise, delivery_address, status
care_hitl_approvals	id (PK) IDX: (status, clinician_id)	triage_note_id, clinician_id, approval_status, digital_signature

4. Executive Agile Roadmap & Delivery Matrix (17 Epics)

Below is a high-level summary of the 17 Agile Epics and 34 Actionable Tickets for scaling, developing, validating, and testing Studentkare. The complete ticket set is exported in **studentkare_agile_roadmap.xlsx**.

Epic Key	Epic Title & Scope	Tickets	Est. SP	Target Sprint
EPIC-SCALE	Infrastructure & Data Scaling	4 Tickets	13 SP	Sprint 1 - 2
EPIC-MONITOR	APM, OpenTelemetry & Grafana Monitoring	2 Tickets	8 SP	Sprint 1
EPIC-QA-TEST	Vitest, Pytest & Playwright QA Suite	3 Tickets	13 SP	Sprint 1, 4
EPIC-BRANDING	Logo Suite & Impilo Pearl Design Tokens	2 Tickets	5 SP	Sprint 1 - 2
EPIC-ANALYTICS	PostHog Behavioral Telemetry & Funnels	1 Ticket	3 SP	Sprint 2
EPIC-FIREBASE	Firebase FCM Push Notifications & Firestore	2 Tickets	10 SP	Sprint 2 - 3
EPIC-WORKFLOW	Follow-up Engine & Critical Vitals Escalations	2 Tickets	10 SP	Sprint 3
EPIC-AGGREGATOR	Wearable Aggregator & ABDM FHIR Vault	2 Tickets	16 SP	Sprint 3
EPIC-DEV-AI	Multi-Language Voice Scribe & Acoustics	2 Tickets	10 SP	Sprint 2
EPIC-DATA-ANALYSIS	Outbreak Analytics & Predictive Risk Scoring	2 Tickets	10 SP	Sprint 4
EPIC-MOBILE-PWA	Offline-First IndexedDB & Background Sync	2 Tickets	8 SP	Sprint 2
EPIC-WEBRTC	WebRTC HD Video & Interactive Whiteboard	2 Tickets	13 SP	Sprint 3
EPIC-CLAIMS	Drools Claims Engine & NHA HCX Gateway	2 Tickets	16 SP	Sprint 3 - 4
EPIC-HA-DR	DB Replication, WAL Archiving & DR Drill	2 Tickets	10 SP	Sprint 2, 4
EPIC-MULTI-TENANT	Tenant RLS Partitioning & University SSO	2 Tickets	13 SP	Sprint 2
EPIC-SPORTS-REHAB	AI Ergonomics Tracker & Sports Physio Rehab	2 Tickets	8 SP	Sprint 2 - 3